Oracle separation of error-prone easiness and errorless average-case hardness of NP

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Editorial note. Faithful editable TeX transcription of the two core average-case separation clauses of author Theorem4, complete masked-oracle construction, Theorem15/Lemmas16-17 with full AppendixA, Theorem19 and Corollaries21-22 with all proof continuations, necessary definitions and complete bibliography. Original author mathematical expressions retained as published. Remaining meta-complexity/learning/cryptographic consequence clauses not separately counted. Standalone typography and long-formula line breaks adapted; no new proofs or mathematical corrections.

Allowed standard prerequisites (editorial).

- Baby switching lemma and pseudorandom generation from a random function oracle. Author [14] switching lemma and exact Theorem18 citing Zimand[45] retained as standard prior results.
- AIPRG-to-hitting-set transformation and Borel-Cantelli lemma. Author invokes [34, Lemma18] for the standard primitive conversion; every local oracle proof and errorless-hardness reduction is retained.

Author notation and average-case conventions (Section 3). For each $n \in \mathbb{N}$, let $[n] = \{1, \dots, n\}$. For each $x \in \{0,1\}^n$ and $i \in [n]$, we let x_i denote the i-th bit of x and $x_{\leq i}$ denote $x_1 \circ \cdots \circ x_i$. For a distribution D, we write $x \leftarrow D$ to refer to a random sampling x according to D. For a finite set S, we also use the notation $x \leftarrow_u S$ to denote the uniform sampling from S. For each $n \in \mathbb{N}$, we let U_n denote the uniform distribution over $\{0,1\}^n$ or a random variable selected uniformly at random from $\{0,1\}^n$ in context. We use the notation negl to represent a certain negligible function, i.e., for any polynomial $p(n)$, $\mathrm{negl}(n) < 1/p(n)$ for sufficiently large $n \in \mathbb{N}$. For a randomized algorithm A using $r(n)$ random bits on an n-bit input, we use $A(x; s)$ to refer to the execution of $A(x)$ with a random tape s for $x \in \{0,1\}^n$ and $s \in \{0,1\}^{r(n)}$. For any oracles O_0 and O_1, we let $O_0 + O_1$ denote the combination, i.e., for any $b \in \{0,1\}$ and any $x \in \{0,1\}^*$, $(O_0 + O_1)(b \circ x) = O_b(x)$. In this paper, we

assume the basic knowledge of probability theory, including the union bound, Markov's inequality, Hoeffding's inequality, and the Borel–Cantelli lemma. For each $p \in [0,1]$ and set S of variables taking binary values, we define a p-random restriction ρ to S as a partial assignment $\rho : S \to \{0,1,*\}$ (where $*$ represents "unassigned") randomly selected as follows: for each $x \in S$,

$$\rho(x) = \begin{cases} * & \text{with probability } p, \\ 0 & \text{with probability } (1-p)/2, \\ 1 & \text{with probability } (1-p)/2. \end{cases}$$

For every restriction ρ to S and function f defined on S, we let $f|_\rho$ denote the restricted function obtained by applying a partial assignment to f according to ρ.

3.1. Average-Case Complexity. We present the notions in average-case complexity theory. Further backgrounds can be found in a survey [7]. We say that a family $D = \{D_n\}_{n\in\mathbb{N}}$ of distributions, where each D_n is a distribution on $\{0,1\}^n$, is (polynomial-time) samplable if there exists a randomized sampling algorithm S such that the distribution of $S(1^n)$ is identical to D_n for each $n \in \mathbb{N}$. We consider a family of distributions as a single distribution on instances. We define a distributional problem as a pair of a language $L \subseteq \{0,1\}^*$ and a distribution $D = \{D_n\}_{n\in\mathbb{N}}$ on instances. For a standard complexity class C (e.g., NP), we define its average-case extension $\mathsf{Dist}C$ as $\mathsf{Dist}C = \{(L,D) : L \in C, D \text{ is samplable}\}$. We present the errorless average-case easiness. We say that a distributional problem (L,D) has an errorless heuristic algorithm A with failure probability $\epsilon : \mathbb{N} \to (0,1)$ if (1) A outputs $L(x)(:= \mathbf{1}\{x \in L\})$ or $\perp$ (which represents "failure") for every $x \in \mathrm{supp}(D)$, and (2) the failure probability that $A(x)$ outputs $\perp$ over the choice of $x \leftarrow D$ is bounded above by $\epsilon(n)$ for each $n \in \mathbb{N}$. Note that an errorless heuristic algorithm never outputs an incorrect value $\neg L(x)$ for any $x \in \mathrm{supp}(D)$. Then, for every $\epsilon : \mathbb{N} \to (0,1)$, we define a class $\mathsf{Avg}_\epsilon\mathsf{P}$ as a class of distributional problems that have a polynomial-time errorless heuristic algorithm with failure probability $\epsilon(n)$. Furthermore, we say that a distributional problem (L,D) has an errorless heuristic scheme A if A is given an instance $x \in \mathrm{supp}(D)$ and $\epsilon \in (0,1)$ as input and satisfies the condition of an errorless heuristic algorithm with failure probability ϵ. We define a class AvgP as a class of distributional problems that have a polynomial-time errorless heuristic scheme. It is not hard to verify that $\mathsf{AvgP} \subseteq \mathsf{Avg}_{1/p(n)}\mathsf{P}$ for any polynomial $p(n)$. Next, we present the error-prone average-case easiness. We say that a distributional problem (L,D) has an error-prone heuristic algorithm A with failure probability $\epsilon : \mathbb{N} \to (0,1)$ if the failure probability that $A(x) \neq L(x)$ over the choice of $x \leftarrow D$ is bounded above by $\epsilon(n)$ for each $n \in \mathbb{N}$. Note that an error-prone heuristic algorithm may output an incorrect value $\neg L(x)$, but the error probability is bounded above by $\epsilon(n)$. Then, for every $\epsilon : \mathbb{N} \to (0,1)$, we define a class $\mathsf{Heur}_\epsilon\mathsf{P}$ as a class of distributional problems that have a polynomial-time error-prone heuristic algorithm with failure probability $\epsilon(n)$. We also define an error-prone heuristic scheme and the class HeurP in the same manner as the errorless case. We also define classes $\mathsf{AvgP/poly}$, $\mathsf{HeurP/poly}$, $\mathsf{AvgSIZE}[s(n)]$, and $\mathsf{HeurSIZE}[s(n)]$ for each size parameter $s(n)$ in the same manner as above.

Definition 11 (auxiliary-input pseudorandom generator). Let $n, m : \mathbb{N} \to \mathbb{N}$ be polynomials. We say that $G = \{G_z : \{0,1\}^{n(|z|)} \to \{0,1\}^{m(|z|)}\}_{z \in \{0,1\}^*}$ is an auxiliary-input pseudorandom generator (AIPRG) against C if each $G_z(x)$ is polynomial-time computable from (z, x), $n(\ell) < m(\ell)$ holds for any $\ell \in \mathbb{N}$, and for any adversary A in C, there exists an infinite subset $Z_A \subseteq \{0,1\}^*$ such that for every $z \in Z_A$,

$$\left|\Pr[A(z, G_z(U_{n(|z|)})) = 1] - \Pr[A(z, U_{m(|z|)}) = 1]\right| < \mathrm{negl}(|z|).$$

When the auxiliary-input z is obvious in context, we write $n(|z|)$ and $m(|z|)$ as n and m, respectively.

Definition 13 (hitting set generator). Let $\ell, m : \mathbb{N} \to \mathbb{N}$ be polynomials. We say that $G = \{G_n\}_{n \in \mathbb{N}}$, where $G_n : \{0,1\}^{\ell(n)} \to \{0,1\}^{m(n)}$ is a hitting set generator (HSG) against C if each G_n is polynomial-time computable, $\ell(n) < m(n)$ holds for each $n \in \mathbb{N}$, and G hits any language recognized by adversaries in C in the following sense: For any adversary A in C, let $L_A = \{L_{A,n}\}_{n \in \mathbb{N}}$ be a language recognized by A, where $L_{A,n} \subseteq \{0,1\}^n$ for each $n \in \mathbb{N}$. Then, for infinitely many $n \in \mathbb{N}$, the following holds:

$$|L_{A,m(n)}| > 2^{m(n)-1} \implies L_{A,m(n)} \cap \operatorname{Im} G_n \neq \varnothing.$$

Theorem 4 (core oracle separation clauses). For any constant $a > 0$, there exists an oracle O relative to which the following hold:

- (Error-prone average-case easiness of NP) $\mathsf{DistNP}^O \subseteq \mathsf{HeurP}^O$.
- (Errorless average-case hardness of NP) $\mathsf{DistNP}^O \not\subseteq \mathsf{AvgSIZE}^O[2^{an/\log n}]$.

4. Oracle Construction. In this section, we formally present the proposed oracle construction. Let $a > 0$ be a parameter.

Construction. $\mathcal{O}_a = \mathcal{F} + \mathcal{A}$, where each oracle is randomly selected according to the following process:

1. Let $t(n) = 2^{an/\log n}$ be the upper bound on the time of nonuniform adversaries and $c = 7a$.
2. Define functions p and $i_{\max}$ as $p(n) = t(n)^{-6}$ and $i_{\max}(n) = \frac{1}{c} \log \log t(n)$. Here, p is a parameter of random restriction, and $i_{\max}$ is the number of applications of random restrictions.
3. For each $n \in \mathbb{N}$, define a set $V_{n,0}$ of variables taking binary values as follows:
$$V_{n,0} = \{F_{z,x,\ell} : z, x \in \{0,1\}^n, \ell \in [n]\}.$$
4. For each $n \in \mathbb{N}$, let $S_{n,0} = \{0,1\}^n$.
5. For each $n \in \mathbb{N}$ and $i \in [i_{\max}(n) - 1]$, we inductively (on i) select a $p(n)$-random restriction $\rho^*_{n,i}$ to $V_{n,i-1}$ and a random subset $S_{n,i} \subseteq S_{n,i-1}$ of

size $p(n) \cdot |S_{n,i-1}|$. Then, we define a restriction $\rho_{n,i}$ to $V_{n,i-1}$ and a subset $V_{n,i} \subseteq V_{n,i-1}$ as follows:

$$\rho_{n,i}(z,x,\ell) = \begin{cases} * & \text{if } z \in S_{n,i}, \\ \rho^*_{n,i}(z,x,\ell) & \text{otherwise,} \end{cases}$$

$$V_{n,i} = \rho_{n,i}^{-1}(*) \left(= \rho_{n,i}^{*-1}(*) \cup \{F_{z,x,\ell} : z \in S_{n,i}\}\right).$$

We also define $\rho_{n,i_{\max}(n)}$ as a full assignment to $V_{n,i_{\max}(n)-1}$ selected uniformly at random. Let $\rho_{n,i} \equiv \rho_{n,i_{\max}(n)}$ for each $i \geq i_{\max}(n)+1$. We use the notation $\rho_{n,\leq i}$ to represent the composite restriction $\rho_{n,1} \cdots \rho_{n,i}$ to $V_{n,0}$ for each n and i.

6. Define $\mathcal{F} = \{F_n\}_{n\in\mathbb{N}}$, where $F_n : \{0,1\}^n \times \{0,1\}^n \times [n] \to \{0,1\}$, as $F_n(z,x,\ell) = \rho_{n,\leq i_{\max}(n)}(z,x,\ell)$.

7. Define $\mathcal{A}$ as follows: On input $(M, x, 1^{T^{2^c}})$, where M is a nondeterministic oracle machine, $x \in \{0,1\}^*$, and $T \in \mathbb{N}$, the oracle $\mathcal{A}(M, x, 1^{T^{2^c}})$ returns 0 or 1 according to the following procedure:
   (a) Let $i_T := \frac{1}{c} \log\log T$.
   (b) Construct a T-DNF ϕ on variables in $V_{n,0}$ representing the execution of $M^{\mathcal{F}+\mathcal{A}}(x)$ in T steps, where the top-most OR corresponds to the nondeterminism on a possible choice of $\mathcal{F}$ (say, F') and an accepting path of $M^{F'+\mathcal{A}}(x)$, and each term performs verification whether M's at most T queries (say, $(z_1,x_1,\ell_1),\ldots,(z_q,x_q,\ell_q)$ for some $q \leq T$) are consistent with the actual choices of $\mathcal{F}$, i.e., for each $i \in [q]$, the term contains F_{z_i,x_i,ℓ_i} if $F'(z_i,x_i,\ell_i) = 1$; otherwise, $\neg F_{z_i,x_i,\ell_i}$ as a literal.
   (c) If $\phi|_{\rho_{1,\leq i_T},\ldots,\rho_{T,\leq i_T}} \equiv b$ for some $b \in \{0,1\}$, then return b, otherwise, return 0.

We can verify that $\mathcal{A}$ is well-defined (i.e., not circular on recursive calls for $\mathcal{A}$) as follows.

Proposition 14. For each input, the value of $\mathcal{A}(M, x, 1^{T^{2^c}})$ is determined only by $\rho_{n,j}$ (equivalently, $\rho^*_{n,j}$ and $S_{n,j}$) for $n \leq T$ and $j \leq i_T$ (remember that $i_T = \frac{1}{c}\log\log T$).

Proof. We show the proposition by induction on T. Remember that, on input $(M, x, 1^{T^{2^c}})$, the oracle $\mathcal{A}$ first makes a T-DNF ϕ based on M independently of the values of $\mathcal{F}$. Suppose that M makes some query $(M', x', 1^{T'^{2^c}})$ to $\mathcal{A}$ for constructing ϕ. Since the length of such a query is at most T, we have $T'^{2^c} \leq T$ and

$$i_{T'} = \frac{1}{c}\log\log T' \leq \frac{1}{c}\log\log T^{1/2^c} = \frac{1}{c}\log\log T - 1 = i_T - 1.$$

By the induction hypothesis, the answer of $\mathcal{A}(M', x', 1^{T'^{2^c}})$ is determined by only $\rho_{n,j}$ for $n \leq T'$ and $j \leq i_T - 1$, and so is ϕ. Then, $\mathcal{A}$ determines the answer by restricting ϕ by $\rho_{n,j}$ for $n \leq T$ and $j \leq i_T$. Therefore, $\mathcal{A}(M, x, 1^{T^{2^c}})$ is determined only by $\rho_{n,j}$ for $n \leq T$ and $j \leq i_T$. □

When the parameter a is clear from the context, we omit the subscript a from $\mathcal{O}_a$.

5. Error-Prone Average-Case Easiness of NP. In this section, we show the error-prone average-case easiness of NP.

Theorem 15. With probability 1 over the choice of $\mathcal{O}_a$, $\mathsf{DistNP}^{\mathcal{O}_a} \subseteq \mathsf{HeurP}^{\mathcal{O}_a}$ holds. First, we introduce several notations. For each choice of $\mathcal{O}$, we define the oracle $\mathcal{A}^*$ in the same manner as $\mathcal{A}$ except we apply $\rho_{1,\le i_T-1}\rho^*_{1,i_T}, \ldots, \rho_{T,\le i_T-1}\rho^*_{T,i_T}$ to ϕ instead of $\rho_{1,\le i_T}, \ldots, \rho_{T,\le i_T}$. Note that $\mathcal{A}^*$ executes a given nondeterministic machine M with access to $\mathcal{A}$ (rather than $\mathcal{A}^*$) to construct the corresponding DNF ϕ. We can verify that $\mathcal{A}^*$ is well-defined (i.e., not circular) in the same manner as Proposition 14. Now, we show that $\mathcal{A}$ and $\mathcal{A}^*$ do not differ considerably.

Lemma 16. For each input $(M, x, 1^{T^{2^c}})$ to $\mathcal{A}$, we have that

$$\Pr_{\mathcal{O}}[\mathcal{A}(M, x, 1^{T^{2^c}}) \neq \mathcal{A}^*(M, x, 1^{T^{2^c}})] = O(T^{-4}).$$

Proof. Let $i := i_T = (1/c)\log\log T$. For all $n \le T$ and $j \le i-1$, we fix $\rho^*_{n,j}$, $S_{n,j}$, and $\rho^*_{n,i}$ arbitrarily; let C_T denote this condition. Notice that the DNF formula ϕ_{C_T} constructed by $\mathcal{A}$ and $\mathcal{A}^*$ is determined only by C_T because all answers to recursive calls for $\mathcal{A}$ are determined by C_T as in Proposition 14. Let $\phi'_{C_T} = \phi_{C_T}|_{\rho_{1,\le i-1},\ldots,\rho_{T,\le i-1}}$. Then, the value of $\mathcal{A}(M, x, 1^{T^{2^c}})$ (resp. $\mathcal{A}^*(M, x, 1^{T^{2^c}})$) is determined by $\phi'_{C_T}|_{\rho_{1,i},\ldots,\rho_{T,i}}$ (resp. $\phi'_{C_T}|_{\rho^*_{1,i},\ldots,\rho^*_{T,i}}$). For any DNF formula ϕ and a restriction ρ, there are the following three cases: (i) $\phi|_\rho \equiv 0$, (ii) $\phi|_\rho \equiv 1$, or (iii) $\phi|_\rho$ does not become a constant (we write this case as $\phi|_\rho \equiv *$). Following this case analysis, there exist $3^2 = 9$ cases on $(\phi'_{C_T}|_{\rho^*_{1,i},\ldots,\rho^*_{T,i}}, \phi'_{C_T}|_{\rho_{1,i},\ldots,\rho_{T,i}})$. However, since each $\rho_{n,i}$ is a subrestriction of $\rho^*_{n,i}$ (i.e., $\rho_{n,i}$ assigns values only to variables that are also assigned by $\rho^*_{n,i}$), the following 4 cases do not occur: $(0,1)$, $(1,0)$, $(*,0)$, and $(*,1)$. Further, in the cases of $(0,0)$, $(1,1)$, $(*,*)$, and $(*,0)$, the answers by $\mathcal{A}^*$ and $\mathcal{A}$ do not differ because $\mathcal{A}^*$ and $\mathcal{A}$ return 0 in the case of $*$. Thus, we only need to consider the case of $(1,*)$. By the aforementioned argument, the probability in the lemma is expressed as follows:

$$\begin{aligned}&\Pr_{\mathcal{O}}[\mathcal{A}(M, x, 1^{T^{2^c}}) \neq \mathcal{A}^*(M, x, 1^{T^{2^c}})]\\ &= \mathbb{E}_{C_T}\left[\Pr_{S_{1,i},\ldots,S_{T,i}}\left[\phi'_{C_T}|_{\rho_{1,i},\ldots,\rho_{T,i}} \equiv * \;\middle|\; C_T, \phi'_{C_T}|_{\rho^*_{1,i},\ldots,\rho^*_{T,i}} \equiv 1\right]\right].\end{aligned}$$

To bound the probability in the right-hand side for each condition, we consider the case in which $\phi'_{C_T}|_{\rho^*_{1,i},\ldots,\rho^*_{T,i}} \equiv 1$. Then, we can select a term τ in ϕ'_{C_T} such that $\tau|_{\rho^*_{1,i},\ldots,\rho^*_{T,i}} \equiv 1$. For each $n \le T$, define $Z_n \subseteq \{0,1\}^n$ as

$$Z_n = \{z \in \{0,1\}^n : \exists (x,\ell) \in \{0,1\}^n \times [n] \text{ s.t. a variable } F_{z,x,\ell} \text{ is contained in } \tau\}.$$

Since ϕ'_{C_T} is a T-DNF, $|Z_n| \le T$ for each $n \le T$. Furthermore, for any $n \in \mathbb{N}$ such that $i_{\max}(n) \le i-1$, we have that $Z_n = \varnothing$ because all such variables must

be assigned by $\rho_{n,\le i-1}$. If $\phi'_{C_T}|_{\rho_{1,i},\dots,\rho_{T,i}} \equiv *$, then $\tau|_{\rho_{1,i},\dots,\rho_{T,i}} \equiv *$ must hold. This event occurs only if $\bigcup_{n\le T}(S_{n,i} \cap Z_n) \neq \varnothing$ holds. Since each $S_{n,i}$ is selected from $S_{n,i-1}$ uniformly at random, this probability is bounded above by

$$\begin{aligned}
\Pr\left[\bigcup_{n\le T}(S_{n,i}\cap Z_n)\neq\varnothing\right] &\le \sum_{n\le T}\Pr[S_{n,i}\cap Z_n\neq\varnothing] \\
&\le \sum_{n\le T: i_{\max}(n)\ge i} |Z_n|\cdot|S_{n,i}|/|S_{n,i-1}| \\
&\le O(T)\cdot \sum_{n: t^{-1}(T)\le n\le T} p(n) \\
&\le O(T^2\cdot t(t^{-1}(T))^{-6}) = O(T^{-4}).
\end{aligned}$$

Thus, we conclude that

$$\begin{aligned}
&\Pr_{\mathcal{O}}[\mathcal{A}(M,x,1^{T^{2^c}}) \neq \mathcal{A}^*(M,x,1^{T^{2^c}})] \\
&= \mathbb{E}_{C_T}\left[\Pr_{S_{1,i},\dots,S_{T,i}}\left[\phi'_{C_T}|_{\rho_{1,i},\dots,\rho_{T,i}} \equiv * \;\middle|\; C_T, \phi'_{C_T}|_{\rho^*_{1,i},\dots,\rho^*_{T,i}} \equiv 1\right]\right] \le O(T^{-4}).
\end{aligned}$$

□

Furthermore, we can show the average-case easiness of NP under the oracle access to $\mathcal{A}^*$ (instead of $\mathcal{A}$). This part is similar to in [25,20]. Thus, we defer the proof to Appendix A.

Lemma 17 ([25,20]). Let M be a $t_M(n)$-time nondeterministic oracle machine and S be a randomized polynomial-time oracle sampling machine. We assume that $S(1^n)$ takes at most $t_S(n)$ time to generate an instance of length n. Then, the following event occurs with probability 1 over the choice of $\mathcal{O}$: for any $n \in \mathbb{N}$,

$$\Pr_{x\leftarrow S^{\mathcal{O}}(1^n)}[M^{\mathcal{O}}(x) \neq \mathcal{A}^*(M,x,1^{T^{2^c}})] \le O(n^{-4}),$$

where $T = \max\{n^{2^c}, t_M(n)^{2^c}, t_S(n)^{2^c}\}$. Now, we derive the average-case easiness of NP from Lemmas 16 and 17.

Proof of Theorem 15. We consider an arbitrary distributional NP problem (L, D) and assume that L is specified by a $t_M(n)$-time nondeterministic oracle machine M, and D is specified by a randomized $t_S(n)$-time oracle sampling machine S. Let $T = \max\{n^{2^c}, t_M(n)^{2^c}, t_S(n)^{2^c}\}$ as in Lemma 17. We construct an error-prone heuristic algorithm B for (L, D) as follows: On input $x \in \{0,1\}^n$, B queries $(M, x, 1^{T^{2^c}})$ to $\mathcal{A}$ and returns the same answer. In the following, we will verify that the error probability of B (over the choice of $\mathcal{O}$ and $S^{\mathcal{O}}(1^n)$) is bounded above by $O(n^{-4})$ for each input size n. Then, by applying Markov's inequality and the Borel–Cantelli lemma, the error probability of B is bounded above by n^{-2} for all sufficiently large n with probability 1 over the choice of $\mathcal{O}$. Since the number of tuples (M, S) is countable, we can conclude that all distributional NP problems have an error-prone heuristic algorithm with error probability at most n^{-2} with probability 1 over the choice of $\mathcal{O}$. Based on

the argument in [24, Proposition 3], this is sufficient for the statement that $\mathsf{DistNP}^{\mathcal{O}} \subseteq \mathsf{HeurP}^{\mathcal{O}}$. Therefore, it is sufficient to show that, for any $n \in \mathbb{N}$,

$$\Pr_{\mathcal{O},x \leftarrow S^{\mathcal{O}}(1^n)}[M^{\mathcal{O}}(x) \neq \mathcal{A}(M,x,1^{T^{2^c}})] \leq O(n^{-4}).$$

Obviously, this event occurs only if (i) $\mathcal{A}(M,x,1^{T^{2^c}}) \neq \mathcal{A}^*(M,x,1^{T^{2^c}})$ or (ii) $M^{\mathcal{O}}(x) \neq \mathcal{A}^*(M,x,1^{T^{2^c}})$ occur. By Lemmas 16 and 17 and the union bound, we have

$$\begin{aligned}
&\Pr_{\mathcal{O},x \leftarrow S^{\mathcal{O}}(1^n)}[M^{\mathcal{O}}(x) \neq \mathcal{A}(M,x,1^{T^{2^c}})] \\
&\leq \Pr_{\mathcal{O},x \leftarrow S^{\mathcal{O}}(1^n)}[\mathcal{A}(M,x,1^{T^{2^c}}) \neq \mathcal{A}^*(M,x,1^{T^{2^c}})] \\
&\quad + \Pr_{\mathcal{O},x \leftarrow S^{\mathcal{O}}(1^n)}[M^{\mathcal{O}}(x) \neq \mathcal{A}^*(M,x,1^{T^{2^c}})] \\
&= O(T^{-4}) + O(n^{-4}) = O(n^{-4}) + O(n^{-4}) = O(n^{-4}).
\end{aligned}$$

□

A. Proof of Lemma 17. Fix $n \in \mathbb{N}$ arbitrarily. Remember that $T = \max\{n^{2^c}, t_M(n)^{2^c}, t_S(n)^{2^c}\}$. Let $i_T = c^{-1}\log\log T$. We first show that the instance $x \in \{0,1\}^n$ generated by $S^{\mathcal{O}}(1^n)$ is determined by only $\rho_{n',j}$ for $n' \leq T$ and $j \leq i_T - 1$ with probability at least $1 - O(n^{-5})$. Then, we show that $\mathcal{A}^*(M,x,1^{T^{2^c}})$ returns $M^{\mathcal{O}}(x)$ with probability at least $1 - O(n^{-5})$ under the condition that x is determined by only $\rho_{n',j}$ for $n' \leq T$ and $j \leq i_T - 1$. If we assume these, then by the union bound, we have the lemma as

$$\Pr_{\mathcal{O},x \leftarrow S^{\mathcal{O}}(1^n)}[M^{\mathcal{O}}(x) \neq \mathcal{A}^*(M,x,1^{T^{2^c}})] \leq O(n^{-5}) + O(n^{-5}) = O(n^{-5})\ (= O(n^{-4})).$$

Now, we show the first claim. Since $t_S(n) \leq T^{1/2^c}$, the answers of $\mathcal{A}$ to queries made by $S^{\mathcal{O}}(1^n)$ are determined by only $\rho_{n',j}$ for $n' \leq T^{1/2^c}$ and $j \leq i_T - 2$. Under an arbitrary condition on restrictions $\rho_{n',j}$ for $n' \leq T^{1/2^c}$ and $j \leq i_{t_S(n)^{1/2^c}} \leq i_T - 2$, the output $S^{\mathcal{O}}(1^n)$ is determined by only $\rho_{n',j}$ for $n' \leq T$ and $j \leq i_T - 1$ unless S queries $(z,x,\ell) \in \bigcup_{n' \leq T^{1/2^c}} \rho^{*-1}_{n',i_T-1}(*)$ to $\mathcal{F}$. Note that, if $n' \in \mathbb{N}$ satisfies $n' < t^{-1}(T^{1/2^c})$, then we have $i_{\max}(n') < c^{-1}\log\log(T^{1/2^c}) = c^{-1}\log\log T - 1 = i_T - 1$. Thus, $\rho^{*-1}_{n',i_T-1}(*) = \varnothing$. Otherwise, each element in $\rho^{*-1}_{n',i_T-1}(*)$ is selected from V_{n',i_T-2} independently with probability $p(n')$. Since $S^{\mathcal{O}}(1^n)$ accesses to $\mathcal{A}$ at most $T^{1/2^c}$ times, such a conditional probability is bounded above by

$$\begin{aligned}
T^{1/2^c} \max_{t^{-1}(T^{1/2^c}) \leq n' \leq T^{1/2^c}} p(n') &= T^{1/2^c} p(t^{-1}(T^{1/2^c})) \\
&= T^{1/2^c} t(t^{-1}(T^{1/2^c}))^{-6} \\
&= (T^{1/2^c})^{-5} \leq n^{-5},
\end{aligned}$$

where the inequality holds because $T \geq n^{2^c}$. Next, we show the second claim. Under the condition that the given instance $x \in \{0,1\}^n$ is determined by only $\rho_{n',j}$ for $n' \leq T$ and $j \leq i_T - 1$, the T-DNF formula ϕ constructed in

$\mathcal{A}^*(M, x, 1^{T^{2^c}})$ is determined only by $\rho_{n',j}$ for $n' \leq T$ and $j \leq i_T - 1$. Then, applying the restriction $\rho_{n',j}$ for $n' \leq T$ and $j \leq i_T$ under this condition is regarded as a $p(n')$-random restriction to V_{n',i_T-1} for each $n' \leq T$, where we can ignore small n' such that $n' < t^{-1}(T)$ because $i_{\max}(n') < c^{-1} \log\log T = i_T$ for such n'. By applying the switching lemma (particularly, the baby switching lemma) for T-DNF [14], the probability that ϕ does not become a constant is at most

$$\begin{aligned} O\left(T \max_{t^{-1}(T) \leq n' \leq T} p(n')\right) &= O(T \cdot t(t^{-1}(T))^{-6}) \\ &= O(T^{-5}) = O(n^{-5}), \end{aligned}$$

where the last equation holds because $T \geq n^{2^c} \geq n$. Remember that $\mathcal{A}^*$ always returns the correct answer whenever ϕ becomes constant. Therefore, the second claim holds.

6. Errorless Average-Case Hardness of NP. In this section, we show the hardness part of our oracle separation. First, we show the existence of AIPRG relative to $\mathcal{O}_a$. Then we show the other hardness results, including the errorless average-case hardness for NP, as corollaries. We use the following theorem, which shows the existence of PRGs based on a random oracle.

Theorem 18 ([45]). For each $n \in \mathbb{N}$, let $R_n : \{0,1\}^n \to \{0,1\}^n$ be a random function oracle, i.e., R_n is selected uniformly at random from $\{f : \{0,1\}^n \to \{0,1\}^n\}$. There exist a polynomial-time deterministic oracle machine $G^?$ and constants $c \geq 1$ and $b, \epsilon > 0$ satisfying the following: For any $n \in \mathbb{N}$ and $x \in \{0,1\}^{cn}$, $G^{R_n}(x)$ generates a binary string of length $4cn$, and all oracle circuits $C^?$ of size 2^{bn} satisfy that

$$\left| \Pr_{U_{cn}}[C^{R_n}(G^{R_n}(U_{cn})) = 1] - \Pr_{U_{4cn}}[C^{R_n}(U_{4cn}) = 1] \right| \leq 2^{-\epsilon n},$$

with probability at least $1 - 2^n$ over the choice of R_n. Furthermore, the result above is relativized, i.e., the above holds in the presence of an arbitrary oracle O independent of the choice of R_n. Now, we show the existence of AIPRG relative to $\mathcal{O}_a$.

Theorem 19. Let $a > 0$ be an arbitrary constant. With probability 1 over the choice of $\mathcal{O}_a$, there exists an AIPRG $G^{\mathcal{O}_a} = \{G_z^{\mathcal{O}_a}\}_{z \in \{0,1\}^*}$ against $\mathsf{SIZE}^{\mathcal{O}_a}[2^{\epsilon an/\log n}]$ for some absolute constant $\epsilon > 0$, where $G_z^{\mathcal{O}_a} : \{0,1\}^{|z|} \to \{0,1\}^{3|z|}$ for each $z \in \{0,1\}^*$.

Proof. Let $G^?$ and c be the oracle machine and the constant in Theorem 18, respectively. Then, we define an AIPRG $G' = \{G'_z\}_{z \in \{0,1\}^*}$ as $G'^{\mathcal{O}}_z(x) = G^{F_{z'}(x')}_{\leq 3|z|}$, where $x \in \{0,1\}^{|z|}$, $z' = z_{\leq \lfloor |z|/c \rfloor}$, $x' = x_{\leq c|z'|}$, and $F_{z'} : \{0,1\}^{|z'|} \to \{0,1\}^{|z'|}$ is defined as $F_{z'}(y) = \mathcal{F}(z', y, 1) \circ \cdots \circ \mathcal{F}(z', y, |z'|)$. The validity of the truncation is verified as that $|x'| = c|z'| \leq c \cdot |z|/c = |z| = |x|$ and $|G^{F_{z'}}(x')| = 4|x'| = 4c|z'| \geq 4c(|z|/c - 1) \geq 4|z| - 4c \geq 3|z|$ for any z with $|z| \geq 4c$. Let $\epsilon = 1/2c$ and $s(n) = 2^{\epsilon an/\log n}$. We show that G' above is an AIPRG against $\mathsf{SIZE}^{\mathcal{O}}[s(n)]$. Suppose there exists a family $C = \{C_n\}_{n \in \mathbb{N}}$ of

oracle circuits of size $s(n)$ that breaks G', i.e., for any sufficiently large $n \in \mathbb{N}$ and any $z \in \{0,1\}^n$,

$$\left|\Pr_{U_n}[C_n^{\mathcal{O}}(z, G_z'^{\mathcal{O}}(U_n)) = 1] - \Pr_{U_{3n}}[C_n^{\mathcal{O}}(z, U_{3n}) = 1]\right| > \frac{1}{\mathrm{poly}(n)}.$$

Fix $n \in \mathbb{N}$ arbitrarily, and let $n' = \lfloor n/c \rfloor$. We consider an arbitrary choice of $\mathcal{O}$ except for the values of $\rho_{n',i_{\max}(n')}$ (we write this condition as R for convenience). Fix $z \in \{0,1\}^n$ such that $z' = z_{\le n'} \in S_{n',i_{\max}(n')-1}$ arbitrary (where $S_{\cdot,\cdot}$ is the random set in the oracle construction). We refer to such z as a hard index. Since the size of C_n is at most $s(n) = 2^{an/(2c\log n)} \le 2^{a\lfloor n/c\rfloor/\log\lfloor n/c\rfloor} = t(n')$ for sufficiently large n (where t is the time-bound function in the oracle construction), the answers to queries by C_n of the form $\mathcal{A}(M, y, 1^{T^{2^c}})$ do not depend on the values of $F_{z'}$ and are determined by condition R because they are determined only by the restrictions $\rho_{\cdot,j}$ for

$$j \le c^{-1}\log\log T \le c^{-1}\log\log s(n)^{1/2^c} \le c\log\log t(n') - 1 < i_{\max}(n').$$

Now, we consider an arbitrary choice of $\rho_{n',i_{\max}(n')}$ except for the values of $F_{z'}$ and denote this condition by R'. It is not hard to verify that $F_{z'}$ is selected uniformly at random even under the conditions R and R'. Therefore, under the conditions R and R', we can identify the query access to $\mathcal{O}$ by C with the query access to another oracle O' (determined only by R and R') and a random function oracle $F_{z'}$ that are selected independently of O'. For any $n \in \mathbb{N}$, let E_n be an event (over the choice of $F_{z'}$) that there exists a circuit C' of size $2^{bn'}$, where b represents the constant in Theorem 18, such that

$$\begin{aligned}&\left|\Pr_{U_n}[C'^{\mathcal{O}}(G_z'^{F_{z'}}(U_n)) = 1] - \Pr_{U_{3n}}[C'^{\mathcal{O}}(U_{3n}) = 1]\right| \\ &= \left|\Pr_{U_n}[C'^{O',F_{z'}}(G^{F_{z'}}(U_n)) = 1] - \Pr_{U_{3n}}[C'^{O',F_{z'}}(U_{3n}) = 1]\right| > \frac{1}{\mathrm{poly}(n)}.\end{aligned}$$

Then, by Theorem 18 (relative to O'), we have $\Pr_{\mathcal{O}}[E_n \mid R, R'] \le 2^{-\Omega(n)}$. By the Borel–Cantelli lemma, E_n occurs only for finitely many n's with probability 1 over the choice of $\mathcal{O}$ conditioned on R, R' (i.e., the choice of $F_{z'}$). By taking the expectation over R, R', we can show that, with probability 1 over the choice of $\mathcal{O}$, there is no family C' of $2^{bn'}$-size circuits satisfying that for any sufficiently large $n \in \mathbb{N}$, there exists a hard index $z \in \{0,1\}^n$ such that

$$\left|\Pr_{U_n}[C'^{\mathcal{O}}(G_z'^{F_{z'}}(U_n)) = 1] - \Pr_{U_{3n}}[C'^{\mathcal{O}}(U_{3n}) = 1]\right| > \frac{1}{\mathrm{poly}(n)}.$$

However, the circuit C in the assumption violates this statement by embedding a hard index z as auxiliary-input because the size is at most $n + s(n) = O(2^{(a/2c)\cdot n/\log n}) = o(2^{bn'})$. Therefore, we conclude that, with probability 1 over the choice of $\mathcal{O}$, there is no such a circuit C of size $s(n)$, and G' is an AIPRG against $\mathsf{SIZE}[s(n)]$. □

Next, AIPRG implies HSG by regarding the auxiliary-input as a part of the hidden input to HSG. It is not hard to verify the security (refer to [34, Lemma 18] for the formal proof).

Corollary 21. Let $a > 0$ be an arbitrary constant. With probability 1 over the choice of $\mathcal{O}_a$, there exists an HSG $G^{\mathcal{O}_a} = \{G_n^{\mathcal{O}_a}\}_{n\in\mathbb{N}}$ against $\mathsf{SIZE}^{\mathcal{O}_a}[2^{\epsilon an/\log n}]$ for some absolute constant $\epsilon > 0$, where $G_n^{\mathcal{O}_a} : \{0,1\}^{2n} \to \{0,1\}^{3n}$ for each $n \in \mathbb{N}$. Furthermore, the existence of HSGs implies the errorless average-case hardness of NP, as observed in [21] in the context of natural proofs and the average-case complexity of the minimum circuit size problem.

Corollary 22. Let $a > 0$ be an arbitrary constant. With probability 1 over the choice of $\mathcal{O}_a$, $\mathsf{DistNP}^{\mathcal{O}_a} \not\subseteq \mathsf{AvgSIZE}^{\mathcal{O}_a}[2^{\epsilon an/\log n}]$ for some absolute constant $\epsilon > 0$.

Proof. Let $G^{\mathcal{O}}$ and ϵ be the HSG and the constant in Corollary 21, respectively. Then, we define the language $L^{\mathcal{O}}$ as $L^{\mathcal{O}} := \operatorname{Im} G^{\mathcal{O}}$. Obviously, $L^{\mathcal{O}} \in \mathsf{NP}^{\mathcal{O}}$ and $(L^{\mathcal{O}}, \{U_n\}_{n\in\mathbb{N}}) \in \mathsf{DistNP}^{\mathcal{O}}$. Thus, it is sufficient to show that $(L^{\mathcal{O}}, \{U_n\}_{n\in\mathbb{N}}) \notin \mathsf{Avg}_{1/4}\mathsf{SIZE}^{\mathcal{O}}[2^{\epsilon an/(4\log n)}]$. For contradiction, we assume that $(L^{\mathcal{O}}, \{U_n\}_{n\in\mathbb{N}}) \in \mathsf{Avg}_{1/4}\mathsf{SIZE}^{\mathcal{O}}[2^{\epsilon an/(4\log n)}]$. Then, there exists a family $C = \{C_n\}_{n\in\mathbb{N}}$ of $O(2^{\epsilon an/(4\log n)})$-size oracle circuits for $(L^{\mathcal{O}}, \{U_n\}_{n\in\mathbb{N}})$, i.e., for any sufficiently large $n \in \mathbb{N}$,

$$\Pr_{y\leftarrow_u\{0,1\}^{3n}}[C_{3n}^{\mathcal{O}}(y) = \bot] \le 1/4$$

and for each $y \in \{0,1\}^{3n}$,

$$C_{3n}^{\mathcal{O}}(y) \in \{\mathbf{1}(y \in L^{\mathcal{O}}), \bot\}.$$

Note that the size of C_{3n} is at most $O(2^{\epsilon a\cdot 3n/(4\log 3n)})$. Next, we define an adversary C' for $G^{\mathcal{O}}$ as follows: for a given $y \in \{0,1\}^{3n}$ (i.e., the length of seed is $2n$), $C'^{\mathcal{O}}_{2n}$ simulates $C_{3n}^{\mathcal{O}}(y)$, and if C_{3n} returns 1 or $\bot$, then C'_{2n} outputs 0; otherwise (i.e., if C_{3n} returns 0), C'_{2n} outputs 1. Then, based on the aforementioned inequalities, it is not hard to verify that for any sufficiently large $n \in \mathbb{N}$,

$$\begin{aligned}\Pr_{y\leftarrow_u\{0,1\}^{3n}}[C'^{\mathcal{O}}_{2n}(y) = 0] &\le \frac{1}{4} + \frac{|\{G^{\mathcal{O}}(x) : x \in \{0,1\}^{2n}\}|}{|\{0,1\}^{3n}|} \\ &\le \frac{1}{4} + \frac{2^{2n}}{2^{3n}} < \frac{1}{2},\end{aligned}$$

and for any $x \in \{0,1\}^{2n}$, we have $C_{3n}^{\mathcal{O}}(G^{\mathcal{O}}(x)) \in \{1, \bot\}$ and $C'^{\mathcal{O}}_{2n}(G^{\mathcal{O}}(x)) = 0$. Therefore, C' succeeds in avoiding $\operatorname{Im} G^{\mathcal{O}}$, and the size is bounded above by $O(2^{\epsilon an/\log n}) = O(2^{\epsilon a(2n)/\log(2n)})$. This contradicts Corollary 21. Thus, we conclude that $(L^{\mathcal{O}}, \{U_n\}_{n\in\mathbb{N}}) \notin \mathsf{Avg}_{1/4}\mathsf{SIZE}^{\mathcal{O}}[2^{\epsilon an/(4\log n)}]$. □

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